GREATEST COMMON DIVISOR · PRIME FACTORS · MODULAR POWERS

Euler's totient function

$\varphi(n)$

Count the integers that share no prime factor with n, then put the count to work.

1 / FROM A GREATEST COMMON DIVISOR TO A COUNT

Which numbers from 1 to 12 are coprime to 12?

The greatest common divisor is the largest positive integer dividing both numbers. A pair is **coprime** when its gcd is 1. Euler's totient counts the integers from 1 to n that are coprime to n:

$$\varphi(n) = \#\{\, k \in \mathbb{Z} : 1 \le k \le n,\ \gcd(k,n) = 1 \,\}.$$

k	**1**	2	3	4	**5**	6	**7**	8	9	10	**11**	12
$\gcd(k,12)$	**1**	2	3	4	**1**	6	**1**	4	3	2	**1**	12

Since $12 = 2^2 \cdot 3$, first cross out the six even entries, then remove 3 and 9 from those left. The survivors are $1, 5, 7, 11$, so $\varphi(12) = 4$.

The count is the same for remainders $0, \ldots, n-1$: replacing n by 0 leaves the gcd with n unchanged.

2 / COUNT A PRIME POWER

Remove the multiples of one prime

For a prime p, only p itself fails the test among $1, \ldots, p$, so $\varphi(p) = p - 1$. For p^r, the only prime factor is still p. Exactly p^{r-1} entries are multiples of p, so

$$\varphi(p^r) = p^r - p^{r-1} = p^r \left(1 - \frac{1}{p}\right).$$

For $9 = 3^2$, cross out $3, 6, 9$: $\varphi(9) = 6$. For $8 = 2^3$, cross out the four even entries: $\varphi(8) = 4$.

Check the factors. $\varphi(4) = 2$ while $\varphi(2)^2 = 1$. The rule in the next panel requires coprime factors.

3 / JOIN TWO COPRIME FACTORS

Two remainders give independent choices

Suppose $\gcd(a,b) = 1$. Each number from 0 to $ab - 1$ gives a pair of remainders on division by a and by b. If two numbers give the same pair, their difference is a multiple of both a and b, hence of ab. Its magnitude is less than ab, so the difference is zero. With ab numbers and ab pairs, each pair occurs once.

A number is coprime to ab exactly when both remainders are coprime to their own moduli. The counts multiply:

$$\varphi(ab) = \varphi(a)\varphi(b) \qquad (\gcd(a,b) = 1).$$

Thus $\varphi(72) = \varphi(8)\varphi(9) = 4 \cdot 6 = 24$.

4 / COMBINE THE PRIME POWERS

One factor for each distinct prime

Prime powers with distinct bases are coprime. Multiply their totients, then collect the prime powers into n:

$$\boxed{\varphi(n) = n \prod_{\substack{p \mid n \\ p \text{ prime}}} \left(1 - \frac{1}{p}\right)} \qquad \varphi(72) = 72 \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right) = 24.$$

Both 12 and 72 have prime factors 2 and 3, so each keeps the same fraction, $\frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$. The exponents change the count through n; the distinct primes determine the fraction.

5 / USE THE COUNT TO REDUCE POWERS

Euler's theorem: a guaranteed return to 1

Modulo 12, multiplying $1, 5, 7, 11$ by 5 gives the same remainders in a different order: $5, 1, 11, 7$.

If $\gcd(u,n) = 1$, multiplying the coprime remainders by u permutes them: the products stay coprime and u can be cancelled modulo n. Let P be the product of those remainders. Then $u^{\varphi(n)} P \equiv P \pmod{n}$. Since P is coprime to n, cancel it:

$$u^{\varphi(n)} \equiv 1 \pmod{n}.$$

Find 5^{242} modulo 72. Here $\gcd(5, 72) = 1$ and $\varphi(72) = 24$, so $5^{24} \equiv 1 \pmod{72}$. Since $242 = 10 \cdot 24 + 2$,

$$5^{242} = (5^{24})^{10} 5^2 \equiv 25 \pmod{72}.$$

The remainder is 25.

Try $n = 60$. Find $\varphi(60)$, then use it to find the remainder of 7^{33} on division by 60.

Check: $\varphi(60) = 16$, $\gcd(7, 60) = 1$, and $33 = 2 \cdot 16 + 1$, so the remainder is 7.

Adapted from *From GCD to Euler's Totient Function*. Reference: NIST DLMF, §§27.2–3. For $n = 1$, the definition gives $\varphi(1) = 1$.

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